Day 11 Activity: Wrangling the `penguins`

Tuesday, June 2, 2026

Your Name Here

Setup

```
library(tidyverse)
library(palmerpenguins)
```

Today's activity uses the `penguins` dataset again. Glimpse it before you start:

```
glimpse(penguins)
```

```
Rows: 344
Columns: 8
$ species           <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island            <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm    <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
$ bill_depth_mm     <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
$ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
$ body_mass_g       <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
$ sex               <fct> male, female, female, NA, female, male, female, male~
$ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

We'll practice using the six verbs we've learned: yesterday's `group_by()` and `summarise()`, and today's `filter()`, `arrange()`, `select()`, and `mutate()`.

Part 1 — Match the verb

Each task can be done with **one** verb (one task needs two). Name the verb(s) you'd use, without writing any code.

Table 1: Fill in the third column.

#	Task	Verb(s)
1	Keep only the Gentoo penguins	[Your answer.]
2	Add a column of body mass in kilograms	[Your answer.]
3	Sort the penguins heaviest-first	[Your answer.]
4	Keep only the `species`, `island`, and `body_mass_g` columns	[Your answer.]
5	One row per species, each with its average body mass	[Your answer.]
6	Drop the penguins whose `sex` is missing	[Your answer.]

Part 2 — filter()

Write a **single pipeline** for each. Run each one and take a look at the number of rows.

2a. Penguins heavier than 5000 g.

```
# Your code:
```

2b. Adelie penguins that live on Dream island (both conditions).

```
# Your code:
```

2c. Penguins that are Adelie *or* Gentoo. Write it **two ways**, once with `|` and once with `%in%`.

```
# Your code:
```

2d. Debug it. This errors. What's the message, why does it happen, and what's the fix?

```
penguins |>
  filter(species = "Chinstrap")
```

[Your explanation and fix.]

Part 3 — mutate() and arrange()

3a. Add a column `bill_ratio` = bill length ÷ bill depth. Then `arrange()` so the largest ratio is on top. Which species ends up at the top?

```
# Your code:
```

[Which species is on top?]

3b. Add a column `size` that is `"large"` when `body_mass_g > 4000` and `"small"` otherwise. (Hint: `ifelse()`.)

```
# Your code:
```

3c. *Reduction or transformation?* You want the **single heaviest** body mass in the whole dataset. Does that belong in `mutate()` or `summarise()`? Write the pipeline to compute the heaviest body mass, and explain why.

```
# Your code:
```

[Why that verb?]

Part 4 — select()

4a. Keep only `species`, `sex`, and the two bill columns.

```
# Your code:
```

4b. Do the same selection of the two bill columns using a **helper** (`starts_with()` or `ends_with()`) rather than explicitly naming them.

```
# Your code:
```

4c. Select `body_mass_g` but rename it to `mass_g` in the same step.

```
# Your code:
```

Part 5 — Put it together

Answer this in **one pipeline**: *Among penguins with a known sex, what is the average flipper length for each species/sex combination, and how many penguins is each average based on? Show the longest-flippered group first.*

Before you write the code, **predict** how many rows the result has.

Prediction (rows): […]

```
# Your pipeline (filter -> group_by -> summarise -> arrange):
```

Now turn the same filtered data into a plot of `body_mass_g` by `species`, filled by `sex`:

```
# Your plot (remember: switch |> to + when you reach ggplot):
```

In one or two sentences: what does the table or plot show?

[Your answer.]

Part 6 — Readability

The pipeline below works, but it's unreadable. Rewrite it following the following style guidelines: a space before each `|>`, `|>` at the end of the line, each named argument of `summarise()` on its own line indented two spaces, and the closing `)` on its own line.

```
penguins|>filter(!is.na(sex))|>group_by(species,sex)|>summarize(n=n(),avg_mass=mean(body_mass
```

```
# Your restyled version:
penguins|>filter(!is.na(sex))|>group_by(species,sex)|>summarize(n=n(),avg_mass=mean(body_mass
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by species and sex.
i Output is grouped by species.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(species, sex))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 6 x 4
# Groups:   species [3]
  species   sex        n avg_mass
  <fct>     <fct>  <int>    <dbl>
1 Gentoo    male      61    5485.
2 Gentoo    female    58    4680.
3 Adelie    male      73    4043.
4 Chinstrap male      34    3939.
5 Chinstrap female    34    3527.
6 Adelie    female    73    3369.
```

> **Tip**
>
> Once you've done it by hand, try the `styler` package: First, highlight the messy, unreadable version, then hit `Cmd/Ctrl + Shift + P`, type "styler", and pick *"Style selection."* Does it match what you wrote?

Code appendix

```r
library(tidyverse)
library(palmerpenguins)
glimpse(penguins)
# Your code:

# Your code:

# Your code:

penguins |>
  filter(species = "Chinstrap")
# Your code:

# Your code:

# Your code:

# Your code:
```

```
# Your code:

# Your code:

# Your pipeline (filter -> group_by -> summarise -> arrange):

# Your plot (remember: switch |> to + when you reach ggplot):

penguins|>filter(!is.na(sex))|>group_by(species,sex)|>summarize(n=n(),avg_mass=mean(body_mass
# Your restyled version:
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